THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM THREE
032 CHEMISTRY
MARKING SCHEME 2026

SECTION A (16 MARKS)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	B	C	C	B	D	C	B	C	C

(01@ = 10 marks)

2.

LIST A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
LIST B	B	E	C	D	A	F

(01@ = 06 marks)

SECTION B (54 MARKS)

3. (a)

(i) $Ca(OH)_{2(aq)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + 2H_2O_{(l)}$ **(2 marks)**

(ii) Neutralisation reaction**(1 mark)**

(iii) Calcium chloride**(1 mark)**

(b)

(i) Any two:**(2 marks)**

- Effervescence/bubbling occurs.
- A gas is evolved.
- The solid carbonate gradually dissolves.

(ii) $Na_2CO_{3(aq)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$ **(2 marks)**

(iii) CO_2 **(1 mark)**

4.

(i) X:group I period 3 **(1 mark)**
Y: group VII period 3 **(1 mark)**

(ii) X **(1 mark)**

(iii) Y **(1 mark)**

(iv) Z has a complete outermost electron shell / stable electronic configuration.**(2 marks)**

(v) Ionic/electrovalent bond. **(1 mark)**

(vi) X loses one electron to form X+, while Y gains one electron to form Y−; the oppositely charged ions attract. **(2 marks)**

5.

(i) Standard solution: A solution whose chemical concentration is precisely known

End point: The point in a titration at which the indicator shows a permanent colour change, indicating that the reaction has reached the required completion. **(2 marks)**

Given:

- Conc HCl =0.100 mol/dm^3
- Vol HCl =20.0 cm^3
- Vol NaOH =25.0 cm^3

$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$ **(1 mark)**

(ii) Moles of HCl used in the titration.

Convert volume:

20.0 cm^3 = 20/1000 dm^3 **(1 mark)**

= 0.0200 dm^3

n = Con × Vol

n =0.100 × 0.0200 **(1 mark)**

n(HCl) = 0.00200 mol

(iii) Moles of NaOH that reacted.

From:

HCl:NaOH=1:1 **(1 mark)**

Therefore:

n(NaOH)=0.00200 mol **(1 mark)**

(iv) Concentration of the NaOH solution in mol/dm^3 (Molarity)

Vol NaOH = 25.0 cm^3 = 0.0250 dm^3 **(0.5 mark)**

Molarity = mole per 1 dm^3

If 0.0250 dm^3 = 0.00200 mol**(0.5 mark)**

1 dm^3 = ? moles

= 0.00200/0.02500 **(0.5 mark)**

Molarity NaOH =0.0800 mol/dm^3 **(0.5 mark)**

6.

(i) Sample A : Soft water **(0.5 mark)**

Sample B: hard water. **(0.5 mark)**

(ii) Removal of temporary hardness by boiling

Boiling decomposes soluble calcium hydrogencarbonate, forming insoluble calcium carbonate, which can be removed.**(1.5 marks)**

$Ca(HCO_3)_{2(aq)} \rightarrow CaCO_{3(s)} + CO_{2(g)} + H_2O_{(l)}$ **(1.5 marks)**

(iii) Removal of permanent hardness using washing soda — 2 marks

Washing soda, Na_2CO_3, supplies carbonate ions which react with dissolved calcium ions to form insoluble calcium carbonate.**(1.5 marks)**

$Ca^{2+}_{(aq)} + CO_3^{2-}{}_{(aq)} \rightarrow CaCO_{3(s)}$ **(1.5 marks)**

(iv) Two disadvantages of hard water —

Any two: **(2 marks)**

- Wastes soap.

- Forms scum with soap.
- Produces scale in boilers and kettles.
- Blocks pipes.
- Reduces efficiency of heating systems.

7.

Given:
Mass of Mg = **4.8 g**
$Ar_{(Mg)} = 24$
Write the equation:
$Mg + 2HCl \rightarrow MgCl_2 + H_2$ **(1 mark)**

(i) The number of moles of magnesium used. (2 marks)

n = m/Mr **(0.5 mark)**

n = 4.8/24 **(0.5 mark)**

= 0.20 mol **(0.5 mark)**

(ii) The volume of hydrogen gas produced at STP.

Moles of hydrogen:
From: $Mg + 2HCl \rightarrow MgCl_2 + H_2$
$Mg:H_2$ =1:1 **(0.5 mark)**
Therefore:
$n(H_2) = 0.20$ mol **(0.5 mark)**

Volume of hydrogen:
1 mol = 24 dm^3 **(0.5 mark)**

0.20 mol = ?

$V = 0.20 \times 24$ **(0.5 mark)**

V=4.8dm3 **(0.5 mark)**

(iii) Observations
Any two: **(2 marks)**
- Effervescence/bubbling occurs.
- Hydrogen gas is evolved.
- Magnesium gradually dissolves.
- Heat may be produced.
- The reaction produces a colourless gas.

(iv) Hydrochloric acid is in excess because the question states that the magnesium reacts with excess hydrochloric acid.**(2 marks)**

8. (a) A Form Three student wants to investigate whether increasing temperature affects the rate at which sugar dissolves in water. State the following:
(i) To investigate the effect of temperature on the rate/time taken for sugar to dissolve in water.**(1 mark)**
(ii) Temperature of water**(1 mark)**
(iii) Time taken for sugar to dissolve **(1 mark)**
(b) Six appropriate steps, 1 mark each:**(6 marks)**
(i) Identify/state the problem.

(ii) Formulate a hypothesis.
(iii) Plan/design the experiment.
(iv) Collect observations/data. /Record results.
(v) Analyse/interprete the results.
(vi) Draw a conclusion.
(vii) Communicate/report findings.

SECTION C (30 MARKS)

9.

(i) Introduction: **(01½ marks)**
- Renewable energy: energy that can be replenished within a short period after use

(ii) Examples of renewable sources of energy
- Solar energy: energy obtained from sunlight **(Any six 0½ marks = 03 marks)**
- Biomass: energy obtained from organic matter
- Biogas: A fuel gas obtained from fermentation of organic waste
- Geothermal energy: energy obtained by tapping the heat from underneath the earth
- Wind energy: Energy harnessed from windmill
- Water power: obtained from hydroelectric plants, tidal stream energy, wave energy or ocean thermal energy.
- Nuclear energy: energy obtained from changes that occurs in the nuclei of atoms

(iii) Adverse consequences of not using renewable energy sources
- Prolonged drought spells and hence famine.
- Drastic change in rainfall patterns:
- Gobal warming and climate change:
- Increased soil erosion and rapid depletion of soil nutrients
- Increased aridity and desertification. **(Any six 01½ marks = 09 marks)**
- Loss of valuable species of economic or medicinal value.
- Broken food chain and reduced ecosystem stability:
- Destruction of animal habitats and shelters:
- Extinction of animal, microbial and plant species
- Loss of biodiversity.

(iv) Conclusion: Any relevant. **(01½ marks)**

Total: 15 marks

10. Reaction:

$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$

(a) Relative formula mass of $CaCO_3$ — **(2 marks)**

Mr =40+(12)+(3×16) =40+12+48 Mr =100

(b) Moles of $CaCO_3$ — **(3 marks)**

n = m/Mr m n = 10.0/100

$n(CaCO_3) = 0.100$ mol

(c) Moles of evolved CO_2 — **(2 marks)**

From:

$CaCO_3 \rightarrow CaO + CO_2$

Mole ratio $CaCO_3:CO2 = 1:1$.

Therefore:

$n(CO_2) = 0.100$ mol

(d) Volume of CO_2 at RTP — **(3 marks)**

Using molar gas volume at RTP **24 dm³ mol⁻¹**:

$V = n \times 24$

$V = 0.100 \times 24$

$V = 2.40\ dm^3$

(e) Mass of CO_2 —**(3 marks)**

$Mr(CO_2) = 12 + (2 \times 16) = 44$

$m = n \times Mr$

$M = 0.100 \times 44$ $m = 4.40$ g

(f) Evidence that reaction is complete — **(2 marks)**

Any two:

(i) No further gas is evolved on continued heating.

(ii) Limewater test shows no further increase in the amount of CO_2 produced.

(iii) The mass of the residue becomes constant.

(iv) Further heating produces no additional change.

Total: 15 marks

11. (a) A student is given a mixture containing iron filings, sand and sodium chloride.

(i) Heterogeneous mixture. Reason: Its components are not uniformly distributed and can be physically distinguished/separated. **(2 marks)**

(ii) Separation procedure **(6 marks)**

Step 1: Use a magnet to remove the iron filings.

Step 2: Add water to the remaining sand and sodium chloride and stir so that sodium chloride dissolves.

Step 3: Filter the mixture to separate the insoluble sand from the sodium chloride solution.**mark**

Step 4: Wash the sand with a little water and dry it.

Step 5: Evaporate/concentrate the filtrate to recover sodium chloride.

Step 6: Dry the recovered sodium chloride crystals.

(b) A water sample contains dissolved calcium hydrogencarbonate.

(i) Temporary hardness **(1 mark)**
(ii) Removing by boiling. **(3 marks)**

Boiling decomposes calcium hydrogencarbonate to form insoluble calcium carbonate.

$Ca(HCO_3)_{2(aq)} \rightarrow CaCO_{3(s)} + CO_{2(g)} + H_2O_{(l)}$

(c) Any three differences. **(3 marks)**

(i) Physical change does not form a new substance; chemical change forms new substance(s).
(ii) Physical change is generally reversible; chemical change is generally difficult to reverse.
(iii) Physical change usually affects state, shape or size; chemical change affects chemical composition.
(iv) Original properties are generally retained in a physical change; new properties appear in a chemical change.

Total = 15 marks